WEEK 1 · LESSON 1 OF 24 · HANDOUT

# Why Python for AI — and Your First Program

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=fWjsdhR3z3c — open the lesson online to watch inside the course.

**Your goal in this lesson:** Explain why Python is the main language of AI, open a free Google Colab notebook, and run your first lines of code.

## Why Python?

Almost every AI tool you have heard of — from ChatGPT's training code to the crop-disease apps built in Africa — is built mostly with **Python**. There are four simple reasons:

1. **It reads almost like English.** `if price > 5000:` means exactly what it says.
2. **Huge free libraries.** Other people have already written the hard parts: NumPy for maths, Pandas for tables, Matplotlib for charts, scikit-learn and PyTorch for machine learning.
3. **One language for everything** — data cleaning, charts, models and even web apps.
4. **A big community.** Almost any error message you meet has been answered online.

You do not need to become a professional programmer to work in AI. But being able to **read and change** simple Python is one of the most valuable skills in this field.

## Your workspace: Google Colab

You do not need to install anything. **Google Colab** is a free website that runs Python for you on Google's computers. It works on a laptop and, for small exercises, even on a smartphone browser.

1. Go to **colab.research.google.com** and sign in with a Google (Gmail) account.
2. Click **New notebook**.
3. You will see a box called a **cell**. Type code in it.
4. Press **Shift + Enter** (or the ▶ button) to run the cell.

A notebook mixes **code cells** and **text cells**, so you can write notes beside your code. Colab saves your notebooks in your Google Drive. NumPy, Pandas and Matplotlib are already installed.

## Your first program

Type this in a cell and run it:

```
print("Muraho! I am learning Python for AI.")
```

`print()` shows something on the screen. The text inside the quotes is called a **string**.

Now try some maths:

```
print(2 + 3)
print(7 * 6)
print(100 / 8)
print(2 ** 10)
```

`**` means "to the power of", so `2 ** 10` is 1024.

## Comments

A line that starts with `#` is a **comment**. Python ignores it — it is a note for people.

```
# Price of one litre of milk in RWF
print(800 * 3)   # three litres
```

Good comments explain **why**, not just what. You will thank yourself later.

## Errors are normal

Type this on purpose:

```
print("Hello"
```

Python shows a **SyntaxError**, because the closing bracket is missing. Every programmer sees errors every day. Read the **last line** of the error first: it usually tells you what went wrong. Fix it and run again.

## Key terms

- **Python** — a programming language that is easy to read and widely used for AI and data.
- **Notebook** — a document that mixes code, results and notes (Colab, Jupyter).
- **Cell** — one box of code or text in a notebook.
- **String** — text inside quotes, such as `"Kigali"`.
- **Comment** — a note after `#` that Python ignores.
- **Error message** — Python's explanation of what went wrong; read the last line first.

**Remember this:** Python is the language of AI because it is readable and has free libraries for everything. Colab lets you run it free in your browser — and errors are just messages to read and fix.

Name: _____ Date: _____

## Activity

### Task 1

Open a new Colab notebook and name it `math-python-week1`.

### Task 2

Print your name, your town and one reason you want to learn AI — on three lines.

### Task 3

Calculate how many hours are in one year ( `365 * 24` ) and how many seconds ( `365 * 24 * 60 * 60` ).

### Task 4

Make one error on purpose, read the last line of the message, then fix it.

## Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Why is Python the main language for AI?

   A. It is the oldest language
   B. It only runs on phones
   C. It is readable and has many free libraries for maths, data and machine learning
   D. It needs no computer

2. What is Google Colab?

   A. A phone game
   B. A type of spreadsheet
   C. A free website that runs Python notebooks in your browser
   D. A paid antivirus

3. How do you run a cell in Colab?

   A. Close the tab
   B. Press Shift + Enter or the ▶ button
   C. Type RUN in capital letters
   D. Restart the computer

4. What does `2 ** 3` give?

   A. 8
   B. 5
   C. 23
   D. 6

5. What does Python do with a line that starts with `#`?

   A. Prints it in red
   B. Runs it twice
   C. Ignores it — it is a comment
   D. Deletes the notebook

6. When you get an error, what should you read first?

   A. Nothing — restart Colab
   B. The page title
   C. The last line of the error message
   D. The first word only

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